

Algebraic Curves

Chapter 8.5. Canonical Divisors

Let $\mathbb{C}$ be a projective curve and $\mathbb{X}$ its nonsingular model. Denote $\mathbb{K} := \mathbb{k}(\mathbb{C}) = \mathbb{k}(\mathbb{X})$ their function field. Let $\Omega := \Omega_{\mathbb{k}}[\mathbb{K}]$ the module of differentials and call its elements ‘differentials’ on $\mathbb{C}$ or $\mathbb{X}$.

Let $\mathbf{p} \in \mathbb{X}$ be a place and define the ‘order’ of the differential $\omega \in \Omega$ as,

$$\text{ord}_{\mathbf{p}}[\omega] = \text{ord}_{\mathbf{p}}[f]$$

Where we fix a uniformizing parameter t of $\mathcal{O}_{\mathbf{p}}[\mathbb{X}]$ and write $\omega = f dt$ for any $f \in \mathbb{K}$. To see that this is well-defined let s be another uniformizing parameter and $f dt = g ds$. By PROPOSITION 7. we see that $f/g = ds/dt \in \mathcal{O}_{\mathbf{p}}[\mathbb{X}]$ and $g/f = dt/ds \in \mathcal{O}_{\mathbf{p}}[\mathbb{X}]$, so $\text{ord}_{\mathbf{p}}[t] = \text{ord}_{\mathbf{p}}[s]$ necessarily. Similar define the divisor of ω to be,

$$\text{div } \omega = \sum_{\mathbf{p} \in \mathbb{X}} \text{ord}_{\mathbf{p}}[\omega] \mathbf{p}$$

We show that this is well-defined in PROPOSITION 8. The divisor $\text{div } \omega$ is called a ‘canonical divisor’. If $\omega' \in \Omega$ is another divisor then $\omega' = f \omega$ for some $f \in \mathbb{K}$, and hence $\text{div } \omega' = \text{div } f + \text{div } \omega$, so $\text{div } \omega' \sim \text{div } \omega$. In the other direction if $\text{div } \omega' \sim \text{div } \omega$, then $\text{div } \omega' = \text{div } f + \text{div } \omega = \text{div}[f \omega]$. Therefore canonical divisors form a coset under linear equivalence and in particular have the same degree.

PROPOSITION 8.

Let $\mathbb{C}$ be a plane curve of degree $n \geq 3$ whose singularities are all ordinary. Define,

$$E = \sum_{\mathbf{q} \in \mathbb{X}} (r_{\mathbf{q}} - 1) \mathbf{q}$$

Let $\mathbb{G}$ be any plane curve of degree $n - 3$. Then $\text{div } \mathbb{G} - E$ is a canonical divisor.

Proof.

By a projective transformation w.l.o.g. let, $Z \cdot \mathbb{C} = \sum_{i=1}^n \mathbf{p}_i$ where $\mathbf{p}_i$ are all distinct and $[1 : 0 : 0] \notin \mathbb{C}$ with no tangent of $\mathbb{C}$ at a singularity passing through $[1 : 0 : 0]$. Define

covers $x := X/Z, y := Y/Z$ and denote $\{F = 0\} = \mathbb{C}$. Let f denote F whose coordinates are from this affine cover. Again define,

$$E_m = m \sum_{i=1}^n \mathbf{p}_i - E$$

Let $\omega := dx$. Given the divisors $\operatorname{div} \mathbb{G} - E$ and $\operatorname{div} \mathbb{G}$ are linearly equivalent it suffices to show $\operatorname{div} \omega = E_{n-3} + \operatorname{div} \partial f / \partial Y$. Namely as $\partial f / \partial Y = \partial F / \partial Y / Z^{n-1}$ we see that this is equivalent to showing,

$$\operatorname{div} dx - \operatorname{div} \partial F / \partial Y = - \sum_{i=1}^n \mathbf{p}_i - E$$

Now from $f = F = 0$ and taking derivations we have,

$$dx = - \frac{\partial f / \partial Y}{\partial f / \partial X} dy = - \frac{\partial F / \partial Y}{\partial F / \partial X} dy$$

So,

$$\operatorname{ord}_{\mathbf{q}} dx - \operatorname{ord}_{\mathbf{q}} \partial F / \partial Y = \operatorname{ord}_{\mathbf{q}} dy - \operatorname{ord}_{\mathbf{q}} \partial F / \partial X$$

Then let $\mathbf{q}$ be a place centered at $\mathbf{p}_i \in \{Z = 0\} \cap \mathbb{C}$. Thus $y^{-1} = Z/Y$ is a uniformizing parameter of $\mathcal{O}_{\mathbf{p}_i}[\mathbb{X}]$. By taking a derivation of this we get $dy = -y^2 dy^{-1}$, so $\operatorname{ord}_{\mathbf{q}} dy = -2$. Then as $\partial F / \partial X$ does not vanish at $\mathbf{p}_i$ so thus is the order of the l.h.s. of our requisite equality at $\mathbf{q}$, 2. In the other case let $\mathbf{q}$ be a place centered at $\mathbf{p} = [a : b : 1] \in \mathbb{C}$. Given $dx = d(x - a)$ we may w.l.o.g. assume $a = b = 0$. If Y is tangent to $\mathbb{C}$ at $\mathbf{p}$, then $\mathbf{p}$ is not a singularity so x is a uniformizing parameter of $\mathcal{O}_{\mathbf{q}}[\mathbb{X}]$ with $\partial F / \partial Y$ not vanishing at $\mathbf{p}$. Hence $\operatorname{ord}_{\mathbf{q}} dx = \operatorname{ord}_{\mathbf{q}} \partial F / \partial Y = 0$. If Y is not tangent to $\mathbb{C}$ at $\mathbf{p}$ then y is the uniformizing parameter instead at which $\operatorname{ord}_{\mathbf{q}} dy = 0$ and $\operatorname{ord}_{\mathbf{q}} \partial f / \partial X = 1/r_{\mathbf{q}}$. This is all components of the requisite equality. ■

(COROLLARY) If W is a canonical divisor then $\deg W \geq 2g - 2$ and $\ell[W] \geq g$. This follows immediately from choosing $W = E_{n-3}$ and §8.2. COROLLARY 3B.

Question 8.24

PROPOSITION.

If $g > 0$, then $n \geq 3$

Proof.

That is, a curve of positive genus must have degree greater than-or-equal to three. To prove this we show that a line or quadratic cannot have positive genus. Given that our curve has only ordinary singularities, it suffices to consider the formula,

$$g = \frac{(n-1)(n-2)}{2} - \sum_i \frac{r_i(r_i-1)}{2}$$

Where r_i is the multiplicity of the singularity i . A line is entirely smooth so the singularity term is zero. Then as $n = 1$ for a line, hence $g = 0$. Any irreducible conic is smooth, so the singularity term is again zero. Then as $n = 2$ for a conic, hence $g = 0$.

■

Question 8.25

We compute $\operatorname{div} dt$ in the case $\mathbb{X} = \mathbb{P}^1$ and $\mathbb{K} = \mathbb{k}(t) = \mathbb{k}(Y/X)$. We first consider $X \neq 0$ at which $Y/X =: s$ is the uniformizing parameter and,

$$dt = d\left[\frac{Y}{X}\right] = ds$$

So $\operatorname{ord}_{\mathbf{p}} dt = 0$ for all $\mathbf{p}$ here. The last point is $X = 0$ or $[0 : 1]$ at which $X/Y =: s$ is the uniformizing parameter. Observe $t = s^{-1}$. Hence,

$$dt = -s^{-2}ds$$

So $\operatorname{ord}_{[0:1]} dt = -2$. Therefore,

$$\operatorname{div} dt = -2[0 : 1]$$

And indeed as in the corollary, $\deg \operatorname{div} dt = -2$.

Question 8.25

PROPOSITION.

For any $\mathbb{X}$ there exists a curve $\mathbb{C}$ birationally equivalent to $\mathbb{X}$ of degree $n \geq 3$ that has only ordinary singularities.

Proof.

This is exactly QUESTION 7.21.

Question 8.26

PROPOSITION.

For $\mathbb{X} = \mathbb{C} = \{Y^2Z - X(X - Z)(X - \lambda Z) = 0\}$, $x = X/Z$, $y = Y/Z$, and $\mathbb{K} = \mathbb{k}(x, y)$ define,

$$\omega = \frac{1}{y}dx$$

Then $\operatorname{div} \omega = 0$

Proof.

Note $\operatorname{div} \omega = \operatorname{div} dx - \operatorname{div} y$. From QUESTION 8.2 we have,

$$\operatorname{div} y = 1[0 : 0 : 1] + 1[1 : 0 : 1] + 1[\lambda : 0 : 1] - 3[0 : 1 : 0]$$

So it suffices to compute $\operatorname{div} dx$. Namely to show that it equals $\operatorname{div} dy$. For this we must determine the singularities of dx . Namely for this it suffices to observe that if x is a uniformizing parameter at $\mathbf{p}$, then $\operatorname{ord}_{\mathbf{p}} dx = 0$ automatically. This precludes all smooth points of $\mathbb{C}$ and from QUESTION 8.2 only excludes the following points.

On $Y = 1$ we have only $[0 : 1 : 0]$ on the affine cover $0 = Z - X^3 + (\lambda + 1)X^2Z - \lambda XZ^2$. The uniformizing parameter for this is $X =: t$ for,

$$x = \frac{1}{\lambda + t^2} + \frac{(\lambda + 1)t^2}{1 + t^2}$$

Which are power series terms of least exponent -2 . Hence in differentiating we find $\operatorname{ord}_{[0:1:0]} dx = -3$.

On $Z = 1$ we have $[0 : 0 : 1]$, $[1 : 0 : 1]$, and $[\lambda : 0 : 1]$ on the affine cover $0 = Y^2 - X^3 + (\lambda + 1)X^2 - \lambda X$. The uniformizing parameter for all three points is $Y =: t$ for,

$$x = \frac{t^2}{\lambda + (\lambda + 1)X + X^2}$$

Which is a power series term of least exponent 2. Hence in differentiating we find $\operatorname{ord}_{\mathbf{p}} dx = 1$ for all three points $\mathbf{p}$.

Therefore,

$$\operatorname{div} dx = 1[0 : 0 : 1] + 1[1 : 0 : 1] + 1[\lambda : 0 : 1] - 3[0 : 1 : 0]$$

Exactly as like $\operatorname{div} y$, so $\operatorname{div} \omega = 0$ as proposed. ■

Question 8.27

PROPOSITION.

If $g > 0$, then there exists effective canonical divisors.

Proof.

Consider the divisor,

$$E_{n-3} = (n-3) \sum_{i=1}^n 1\mathbf{p}_i - \sum_{\mathbf{q} \in \mathbb{X}} (r_{\mathbf{q}} - 1)\mathbf{q}$$

We saw from PROPOSITION 8. that it was canonical. Note that each $\mathbf{p}_i$ expands as,

$$\begin{aligned} (n-3) \sum_{i=1}^n 1\mathbf{p}_i - \sum_{\mathbf{q} \in \mathbb{X}} (r_{\mathbf{q}} - 1)\mathbf{q} &= (n-3) \sum_{i=1}^k \sum_{\mathbf{q}_{i,j} \in f^{-1}[\mathbf{p}_i]} 1\mathbf{q}_{i,j} - \sum_{\mathbf{q} \in \mathbb{X}} (r_{\mathbf{q}} - 1)\mathbf{q} \\ &= \sum_{\mathbf{q} \in \mathbb{X}} (n-3)\mathbf{q} - \sum_{\mathbf{q} \in \mathbb{X}} (r_{\mathbf{q}} - 1)\mathbf{q} \\ &= \sum_{\mathbf{q} \in \mathbb{X}} (n - r_{\mathbf{q}} - 2)\mathbf{q} \end{aligned}$$

So it suffices to prove that $n - r_{\mathbf{q}} - 2 \geq 0$ everywhere. The contrary could only happen if we have a singularity of multiplicity greater than $n - 2$ at some $\mathbf{q}_0$, i.e. $r_{\mathbf{q}_0} \geq n - 1$. However by our assumption on the genus $g > 0$ we have the inequality,

$$(n-1)(n-2) > \sum_{i=0}^M r_{\mathbf{q}_i} (r_{\mathbf{q}_i} - 1) \geq (n-1)(n-2) + \sum_{i=1}^M r_{\mathbf{q}_i} (r_{\mathbf{q}_i} - 1)$$

Which is a contradiction. Therefore E_{n-3} is indeed effective. ■